

Subject Name: APPLIED CHEMISTRY – II

UNIT 1:

FUELS

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Unit No: 1 Unit name:Fuels

Lecture No: 1

Introduction



Introduction of Fuels

- **Fuel is a substance which when burnt, gives a large amount of heat. Fuel must contain a combustible element like carbon. In the process of combustion the chemical energy of fuel is converted into the heat energy. This heat energy can be used economically for domestic and industrial purposes.**
- **Wood, charcoal, coal, kerosene, petrol, diesel, producer gas, oil gas, water gas etc. are some of the fuels. The main source of fuels are coal and petroleum oils, which are stored fuels available in Earth's crust and are generally called fossil fuels.**



Introduction of Fuels

Definition: A fuel can be defined as a “combustible substance, containing carbon as a main constituent, which on proper burning gives a large amount of heat, which can be used economically for domestic and industrial purposes”.-



Classification of Fuels

There are two type of fuels.

i) Natural or primary fuels.

ii) Derived or secondary fuels.

TYPE	NATURAL (PRIMARY)	DERIVED (SECONDARY)
Solid	Wood, lignite coal , anthracite	Charcoal , coke.
Liquid	Crude petroleum	Fuel oil , diesel, gasoline, kerosene
Gaseous	Natural gas	Producer gas , water gas , coal gas , bio gas .



Characteristics of a Good Fuel

A good fuel should have following characteristics:

- a) High Calorific Value**
- b) Moderate Ignition Temperature**
- c) Low Moisture Content**
- d) Low non-combustible matter content**
- e) Product of combustion should not be harmful**
- f) Low cost**
- g) Easy to transport**



Unit No: 3 Unit name:Applied Chemistry-II

Lecture No: 2

Calorific value (HCV & LCV),Dulong's Formula



Calorific value

Definition: Calorific value of fuel is defined as “ the quantity of heat evolved by the complete combustion of unit quantity (mass or volume) of the fuel in air or oxygen”.

Generally, calorific value of fuel are expressed in two ways:

- a) Gross (Higher) calorific value, represented as GCV or HCV, and**
- a) Net (Lower) calorific value, represented as NCV or LCV.**



Calorific value

- **GCV** can be defined as, “The total amount of heat produced, when unit mass or volume of the fuel has been burnt completely and **the product of combustion have been cooled to room temperature**”.
- **NCV** can be defined as, “ The net heat produced, when a unit quantity of fuel is burnt completely and **the product of combustion are permitted to escape in to the atmosphere**”.



Determination of Calorific Value

Determination of Calorific Value

Theoretically Determination

Dulong's formula for calculating the calorific value from the chemical composition of the fuel may be written as follows:

$$\text{Gross calorific value} = \frac{1}{100} \left[8,080 C + 34,500 \left(H - \frac{O}{8} \right) + 2,240 S \right] \text{ kcal/kg}$$

$$\text{Lower calorific value} = \text{Higher calorific value} - \frac{9}{100} H \times 587 \text{ kcal/kg}$$

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Determination of Calorific Value

Where ,

C = % Carbon

H = % Hydrogen

O = % Oxygen

S = % Sulphur



Unit No: 3 Unit name:Applied Chemistry-II

Lecture No: 3

Analysis of Coal



Analysis of Coal

➤ Coal is a compound of carbon, hydrogen, oxygen, nitrogen and sulphur etc. Along with some inorganic matter. The calorific value of coal depends upon its composition.

➤ In order to determine the quantity of coal, two types of analysis are carried out.

i) Proximate Analysis

ii) Ultimate Analysis.



Analysis of Coal

i) Proximate Analysis-

This method is used for the determination of moisture content, volatile matter content, percentage of ash and percentage of fixed carbon in coal for a particular domestic or industrial use. It is carried out as per standard specification, by using air dried coal sample.



Proximate Analysis

Significance of % of Moisture-

- i) Presence of moisture in the coal lower its quality, since the part of heat produced is used up in evaporating the moisture, thereby the net useful heat is less.**
- ii) So, lesser the moisture content better the quality of coal.**



Proximate Analysis

1. % of Moisture-

- i) About 1 gm of powdered air dried coal sample is weighed in a crucible.
- ii) The crucible is placed inside an electric hot air oven, which is maintained at 105-110°C.
- iii) Then the crucible is allowed to remain in oven for 1 hour.
- iv) Then it is taken out with the help of a pair of tongs.
- v) It is then cooled in a dessicator & weighed.
- vi) Loss in weight is reported as moisture on percentage basis.

$$\% \text{ of moisture} = \frac{\text{Loss in weight}}{\text{Wt. of air dried sample}} \times 100$$



Proximate Analysis

2. % of Volatile Matter-

- i) The moisture free coal sample is left in crucible no.1 is then covered with lid.
- ii) It is then placed in an electric furnace , maintained at $925 \pm 20^\circ\text{C}$.
- iii) The crucible is taken out after 7 minutes of heating.
- iv) The crucible is first cooled in air and then inside a dessiccator and weighed again.
- v) Loss in weight is reported as v.m. on percentage basis.

$$\% \text{ of V.M.} = \frac{\text{Loss in wt. due to removal of V.M.} \times 100}{\text{Wt. of coal sample taken}}$$



Proximate Analysis

Significance of % of Volatile Matter-

- i) The classification of coal depends upon the volatile matter content.
- ii) The volatile matter content in coal is related to the length of the flame smoke forming tendency and the ignition characteristics.
- iii) High volatile matter gives long flame, high smoke and relatively low heating values.
- iv) Hence, lesser the quantity of volatile matter the better the quality of coal.



Proximate Analysis

3. % of Ash -

- i) The residual coal left in crucible 2. is then heated without lid in a muffle furnace at $700 \pm 50^\circ\text{C}$ for 30 minutes.
- ii) The crucible is then taken out. It is cooled in air and then in desiccators and weighed.
- iii) The process of heating, cooling and weighing is repeated till a constant wt. is obtained. The residue is reported as ash on % basis.

$$\% \text{ of ash} = \frac{\text{Wt. of ash left}}{\text{Wt. of coal taken}} \times 100$$



Proximate Analysis

Significance % of Ash -

- i) Ash is non-combustible inorganic matter consist of silica, alumina, iron-oxide and magnesia.
- ii) If ash content is high the ash disposal is a problem. A good fuel should have less ash content.



Proximate Analysis

4. Fixed Carbon-

% of Fixed Carbon = 100 - % of (Moisture + V.M. + ASH)

Significance-

Higher the % of fixed carbon greater is its calorific value and better will be the quality of fuel. Hence higher % of carbon is desirable.



Ultimate Analysis

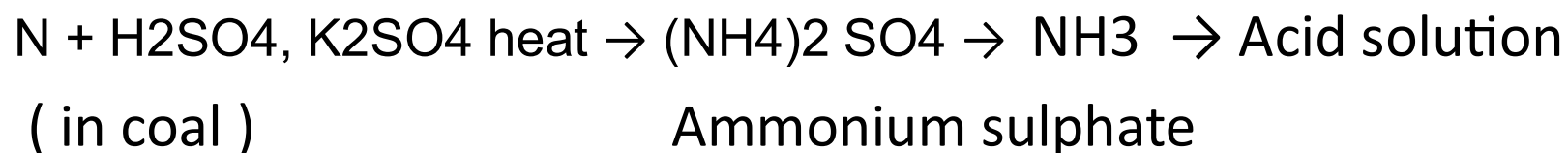
- **Ultimate analysis** encompasses the quantitative determination of % of carbon, hydrogen, nitrogen, sulfur and oxygen within the coal.
- Additionally, specific physical and mechanical properties of coal and particular carbonization properties



ULTIMATE ANALYSIS

Determination of % of nitrogen by kjeldahls method.

- A known weight of powdered air dried coal sample is heated with concentrated H_2SO_4 along with K_2SO_4 catalyst in a kjeldahls flask.
- After the content become clear, it is treated with alkali solution like NaOH or KOH in a round bottom flask. The ammonia liberated is passed in a known volume of standard acid solution.



ULTIMATE ANALYSIS

Determination of % of nitrogen by kjeldahls method.

$$N = \frac{\text{Volume of acid used} \times \text{Normality of acid} \times 1.4}{\text{Weight of coal sample}}$$

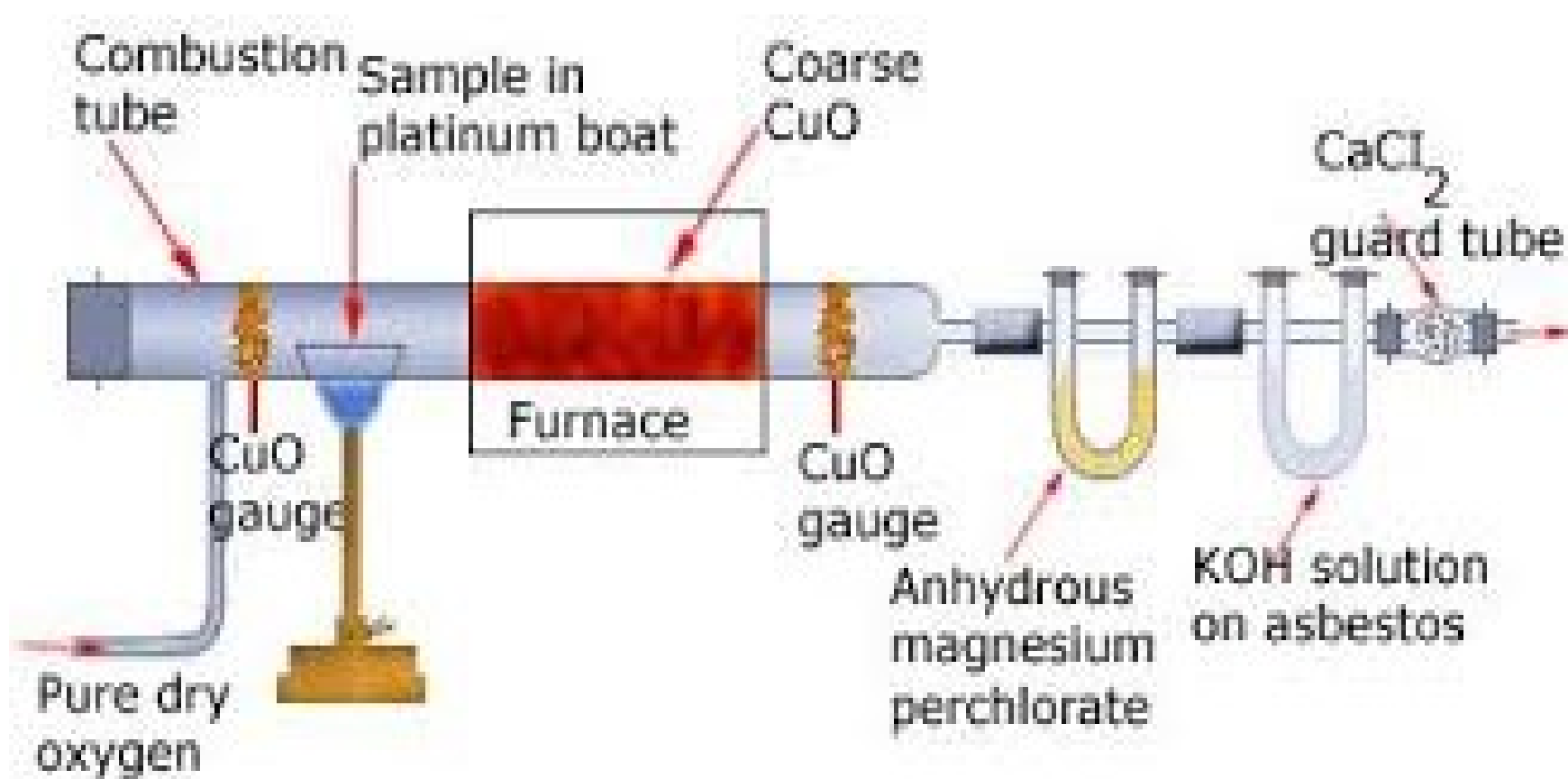
Significance :

Nitrogen does not burn during coal combustion and therefore it has no calorific value. Hence a good quality of coal should have negligible N %.



ULTIMATE ANALYSIS

Determination of % of carbon & hydrogen



ULTIMATE ANALYSIS

Determination of % of carbon & hydrogen

Determination of C & H

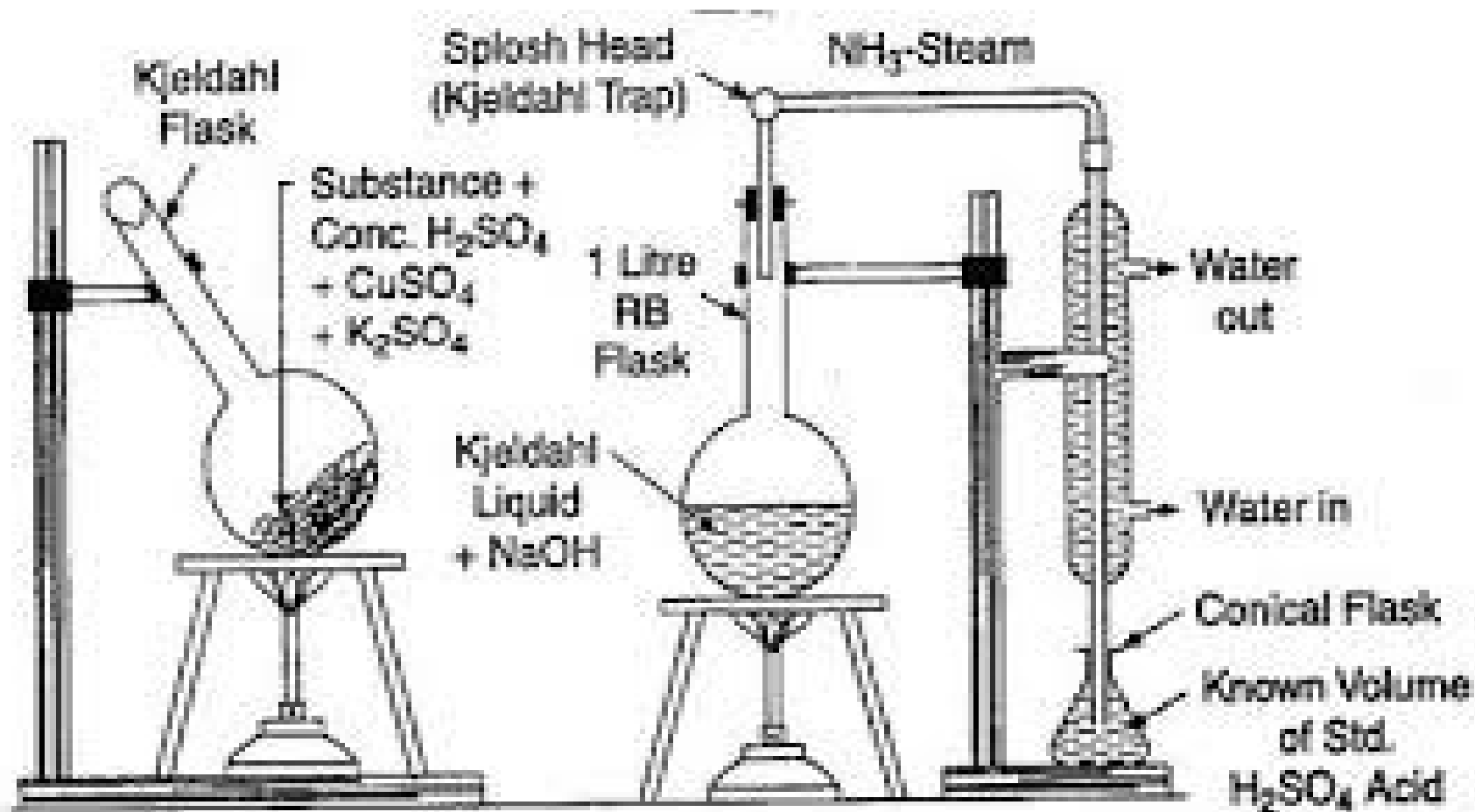
- Accurately weighed coal sample is burnt in a current of oxygen in a combustion apparatus, which is heated to about 350°C .
- Carbon and hydrogen of coal are converted into water vapour and carbon-dioxide. The products of combustion are absorbed in anhydrous CaCl_2 and KOH tubes respectively of known weights.
- After complete absorption of H_2O and CO_2 , the tubes are again weighted.

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ULTIMATE ANALYSIS

Determination of % of nitrogen by kjeldahls method.



ULTIMATE ANALYSIS

Determination of % of carbon & hydrogen

The % of C & H are calculated as follows:

$$C = \frac{\text{Increased in weight of KOH tube}}{\text{Weight of coal sample}} \times \frac{12}{44} \times 100$$

$$H = \frac{\text{Increased in weight of CaCl}_2 \text{ tube}}{\text{Weight of coal sample}} \times \frac{2}{18} \times 100$$



ULTIMATE ANALYSIS

Significance of % of carbon & hydrogen

- Calorific value of fuel is related to its carbon content. Higher the % of carbon, greater is the calorific value & better is the quality of p helps in the ranking of coal & classification of coal.
- Since hydrogen is mainly present in combination with oxygen as water which is not desirable as it lowers the calorific value of fuel, so lesser the % of hydrogen better the quality of coal.



ULTIMATE ANALYSIS

Determination of % of Sulphur

- A Known weight of coal sample is burnt completely in a bomb calorimeter in a stream of O_2 . During burning , sulphur present in the coal gets oxidised & is converted into SO_2 , which on cooling in calorimeter is absorbed by water to form dilute H_2SO_4 .
- Then collect the washings of the Bomb pot in a beaker. Add $BaCl_2$ solution in it.
- Then filter the precipitate of $BaSO_4$, dry it & weigh the ppt of $BaSO_4$.
- From wt. of $BaSO_4$ ppt. % of sulphur calculated.



ULTIMATE ANALYSIS

Determination of % of Sulphur

$$S = \frac{\text{Weight of BaSO}_4 \text{ ppt} \times 32 \times 100}{\text{Weight of Coal Sample} \times 233}$$

Significance : Although sulphur can burn & increase calorific value of coal but it causes SO_x pollution & causes acid rain, corrosion of metallic equipment. Hence, lower the % of sulphur in coal, better is coal quality.



Important formula

Determination of % of Sulphur

$$1. HCV = \frac{1}{100} \times \left[8080 C + 34500 \left(H - \frac{O}{8} \right) + 2240 S \right]$$

$$2. LCV = \left[HCV - \frac{9}{100} H \times 587 \right]$$

$$3. \% \text{ of moisture} = \frac{\text{Loss in wt.of coal}}{\text{Wt.of coal taken}} \times 100$$

$$4. \% \text{ of volatile matter} = \frac{\text{Loss in wt.due to volatile matter}}{\text{Wt.of coal taken}} \times 100$$

$$5. \% \text{ of ash} = \frac{\text{Wt.of ash left}}{\text{Wt.of coal taken}} \times 100$$

$$6. \% \text{ of fixed carbon} = 100 - (\% \text{ Moisture} + \text{Volatile matter} + \text{ash})$$



Important formula

$$7. \% \text{ of carbon} = \frac{\text{Increase in wt. of KOH tube} \times 12 \times 100}{\text{Wt. of coal taken} \times 44}$$

$$8. \% \text{ of hydrogen} = \frac{\text{Increase in wt. of CaCl}_2 \text{ tube} \times 2 \times 100}{\text{Wt. of coal taken} \times 18}$$

$$9. \% \text{ of nitrogen} = \frac{\text{Volume of acid used} (V_1 - V_2) \times \text{Normality of NaOH} \times 1.4}{\text{Wt. of coal taken}}$$

$$10. \% \text{ of Sulphur} = \frac{\text{Weight of BaSO}_4 \text{ ppt} \times 32 \times 100}{\text{Weight of Coal Sample} \times 233}$$

$$11. \% \text{ of } O_2 = 100 - (\% C + \% H + \% N + \% S + \% \text{ash})$$



Unit No: 1 Unit name:Applied Chemistry-1

Lecture No: 4
MINING OF PETROLEUM
REFINING OF PETROLEUM



LIQUID FUEL

MINING OF PETROLEUM

Petroleum (Mineral Oil)

1. Petroleum is natural product generated from organic material by physical & chemical processes inside the earth.
2. Composition : Petroleum is dark greenish brown viscous liquid as found deep in earth's crust, thus known as mineral oil or crude oil. The oil is usually found floating upon a layer of brine & surrounded by natural gases.
3. It is complex mixture of hydrocarbons like straight chain paraffin's, Cycloparaffins, naphthalene's , olefins, aromatic compounds & organic compounds containing Nitrogen, Sulphur, Oxygen, Carbon etc.



LIQUID FUEL

MINING OF PETROLEUM

Petroleum (Mineral Oil)

The average composition of crude petroleum is

Carbon = 75 to 87.1%

Hydrogen = 11.5 to 14.8%

Nitrogen & Oxygen = 0.1 to 0.5%

Sulphur = 0.1 to 3.5%



LIQUID FUEL

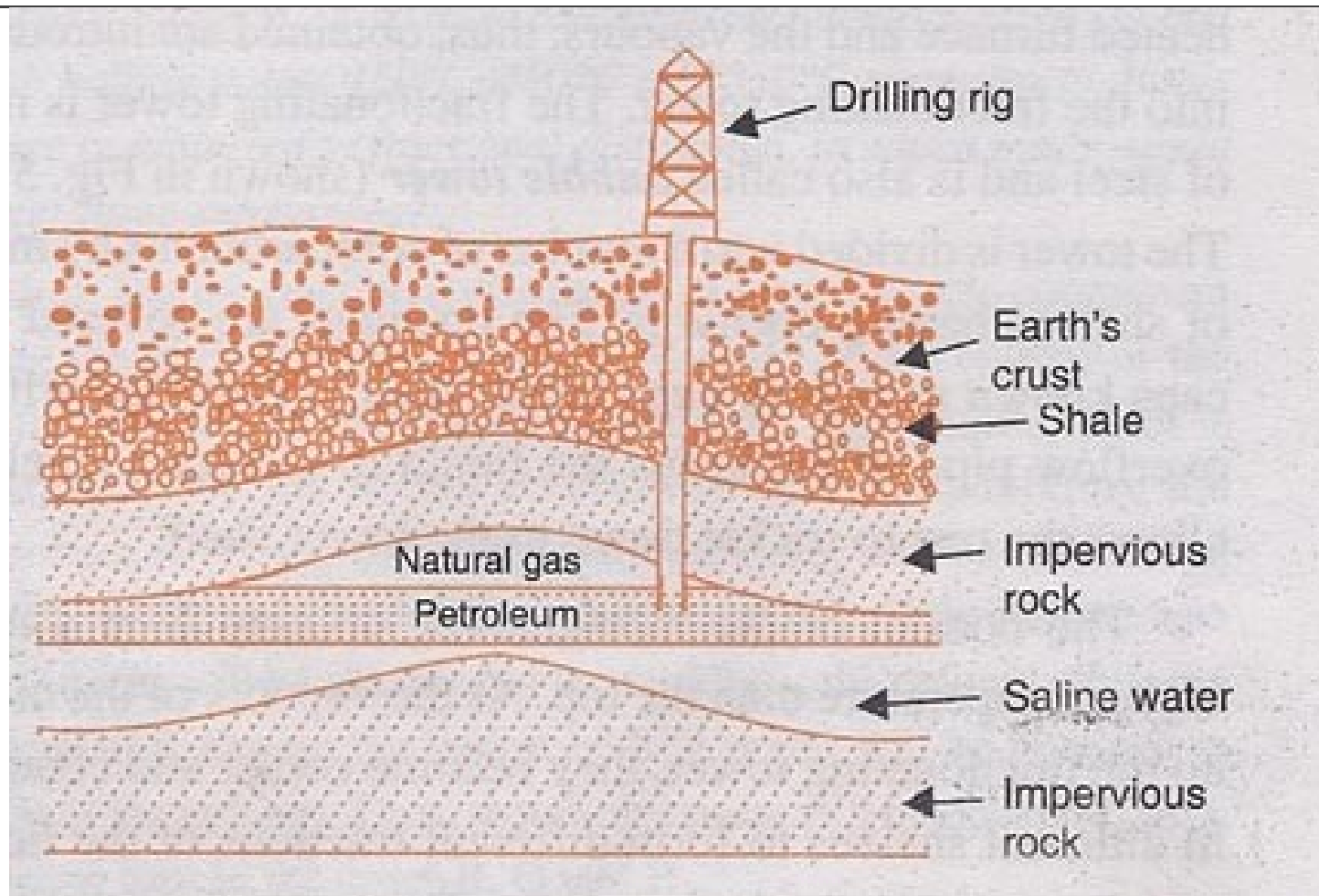
MINING OF PETROLEUM

1. petroleum is obtained by drilling holes in the earth's crust and sinking pipes into it as shown in Fig
2. When the pipe reaches the oil deposit, natural gas comes out with a great pressure and for a time the crude oil comes out by itself due to the pressure of the gas over oil.
3. This is the primary recovery process. Gradually the pressure of the gas decreases. At this stage, the crude oil is extracted with the help of lift pumps. This is the secondary recovery process.



MINING OF PETROLEUM

1.



REFINING OF PETROLEUM

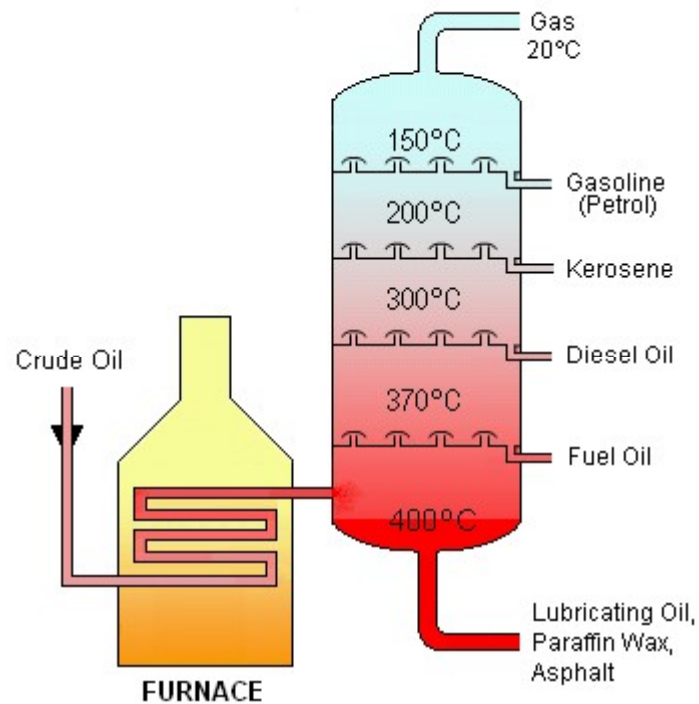


Table Products of fractional distillation of crude oil

Sr. No.	Name of fraction	Boiling range	Composition of hydrocarbon	Uses
1.	Uncondensed gases	Below 40° C	C ₁ to C ₄	Domestic and industrial fuel under 'LPG' name.
2.	Aviation fuel or petroleum ether	40-70° C	C ₅ to C ₇	Fuel for aeroplane, helicopters, as solvent.
3.	Petrol or gasoline	60° C - 120° C	C ₅ to C ₈	Fuel for petrol engines, dry cleaning, as solvent.
4.	Naphtha or solvent spirit	120° C - 180° C	C ₇ to C ₁₀	As solvent and for dry cleaning, for chemicals.
5.	Kerosene	180° C - 250° C	C ₁₀ to C ₁₆	For illumination, domestic fuel, for oil gas and fuel of jet engines.
6.	Diesel	250° C - 320° C	C ₁₅ to C ₁₈	Diesel engine fuel.
7.	Heavy oil (a) Lubricating oils (b) Vaseline (c) Greases (d) Wax	320° C-400° C — — — —	C ₁₇ to C ₃₀ C ₁₈ to C ₂₀ C ₂₀ to C ₂₂ C ₂₂ to C ₂₆ C ₂₆ to C ₃₀	For making petrol by cracking. Lubricating purposes, Cosmetics, medicines, ointments, lubricant. Lubrication. Candles, boot polish, tarpaulin cloth, wax paper, etc.
8.	Residue (a) asphalt (b) petroleum coke	— —	above C ₃₀	Road making, water proofing of roofs. As fuel, for arc light rods.

REFINING OF PETROLEUM

The word petroleum is derived from the Latin word Petra means Rock and Oleum means Oil. Petroleum is a complex mixture of different hydrocarbons. The composition of petroleum varies according to its occurrence. Petroleum in solid form is called ASPHALT.

- Petroleum in liquid form is called CRUDE OIL.
- Petroleum in gaseous form is called NATURAL GAS.
- Petroleum is the most important natural source of fuel.

Petroleum is a complex mixture of hydrocarbons and a number of different compounds whose composition varies according to its place of occurrence.



REFINING OF PETROLEUM

REFINING OF PETROLEUM

- It is the process by which the crude petroleum oil is free from its impurities , separated into various fractions having different boiling points.
- Refining consists of following steps
 1. Separation of water
 2. Removal of Sulphur compounds
 3. Fractional distillation



REFINING OF PETROLEUM

FRACTIONAL DISTILLATION

1.The process of separating a mixture into a series of fractions of different volatilities by means of distillation is known as fractional distillation.

2.In the process of fractional distillation, a mixture of different liquids is evaporated followed by condensation. Different liquids are evaporated according to their boiling point and they are collected in different chambers of distillation tower.



REFINING OF PETROLEUM

FRACTIONAL DISTILLATION OF CRUDE OIL

Generally crude oil contains compounds having boiling points upto 400°C . In this process, crude oil is heated to above 400°C in a pipe. Its vapors are allowed to enter into a fractionating tower having different compartment to collect different volatilities. These are now condensed to liquid and are separated.

- **GASOLINE**
- Gasoline which is commonly known as petrol is a mixture of n-hexane and n-heptane. During the process of fractional distillation, it is obtained in the upper most portion of the tower.



Unit No: 1 Unit name:Fuels

Lecture No: 5

Craking and Types of Cracking



CRACKING

Cracking is the name given to breaking up large hydrocarbon molecules into smaller and more useful bits. This is achieved by using high pressures and temperatures without a catalyst, or lower temperatures and pressures in the presence of a catalyst.



There are two types of cracking.

i) **Thermal cracking**

ii) **Catalytic cracking**



CRACKING

Catalytic cracking

The quality & yield of gasoline produced by cracking can be greatly improved by using a suitable catalyst like $\text{Al}_2(\text{SiO}_3)_3$ or Al_2O_3 at much lower temperature (425- 530c) & pressure (1.5- 5 kg/cm²).

Catalytic cracking done by two methods:

- i) Fixed-bed cracking
- ii) Moving-bed cracking



CRACKING

i) Fixed-bed cracking:

- The heavy oil is passed through a heater at 400-500°C. The bauxite catalyst is mixed with clay & zirconium oxide & packed in catalyst chamber.
- The hot vapours of oil are passed over a fixed bed of catalyst at 425-450°C & 1.5 kg/cm² pressure. About 40% of heavy oil is converted into gasoline & about 2-4% carbon formed gets absorbed in catalyst bed.
- The cracked vapours are then passed through the fractionating column to condense heavy fractions. The vapours are then passed through the cooler where some of the gases are condensed along with gasoline & uncondensed gases move on.



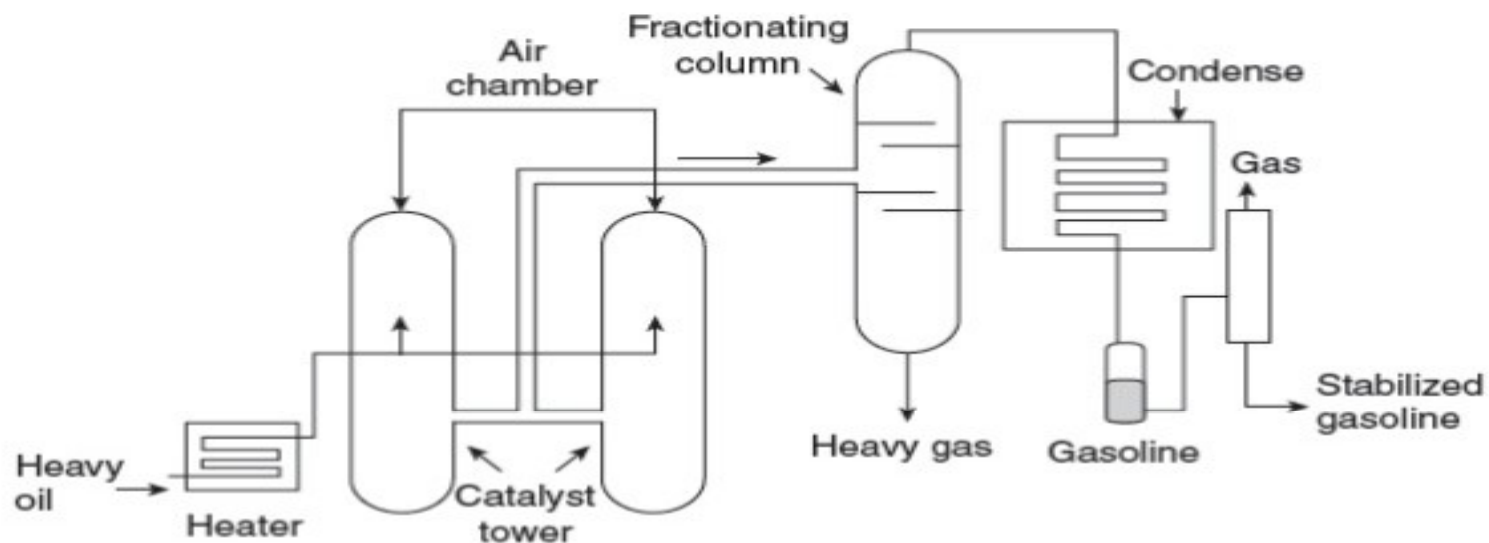
CRACKING

i) Fixed-bed cracking:

- The gasoline containing some dissolved gases then sent to a stabilizer, where the gases are removed & pure gasoline is obtained.
- The catalyst stops functioning after 8 to 10 hours due to the deposition of carbon.



Fixed bed catalytic cracking



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MOVING-BED CRACKING

ii) Moving-bed cracking

- In this process, very finely powdered catalyst is circulated in gas stream which behaves almost as a fluid.
- The oil vapours are cracked in reactor maintained at 530c. & pressure of 5kg/cm². The catalyst dust is removed by passing the cracked vapours through the catalyst dust separator.
- The cracked vapours are then passed through the fractionating column where heavy oil is separated, Then the vapours are passed through the cooler in which the gasoline condense along with some dissolved gases which are removed by passing the gasoline through degasifier.



MOVING-BED CRACKING

ii) Moving-bed cracking

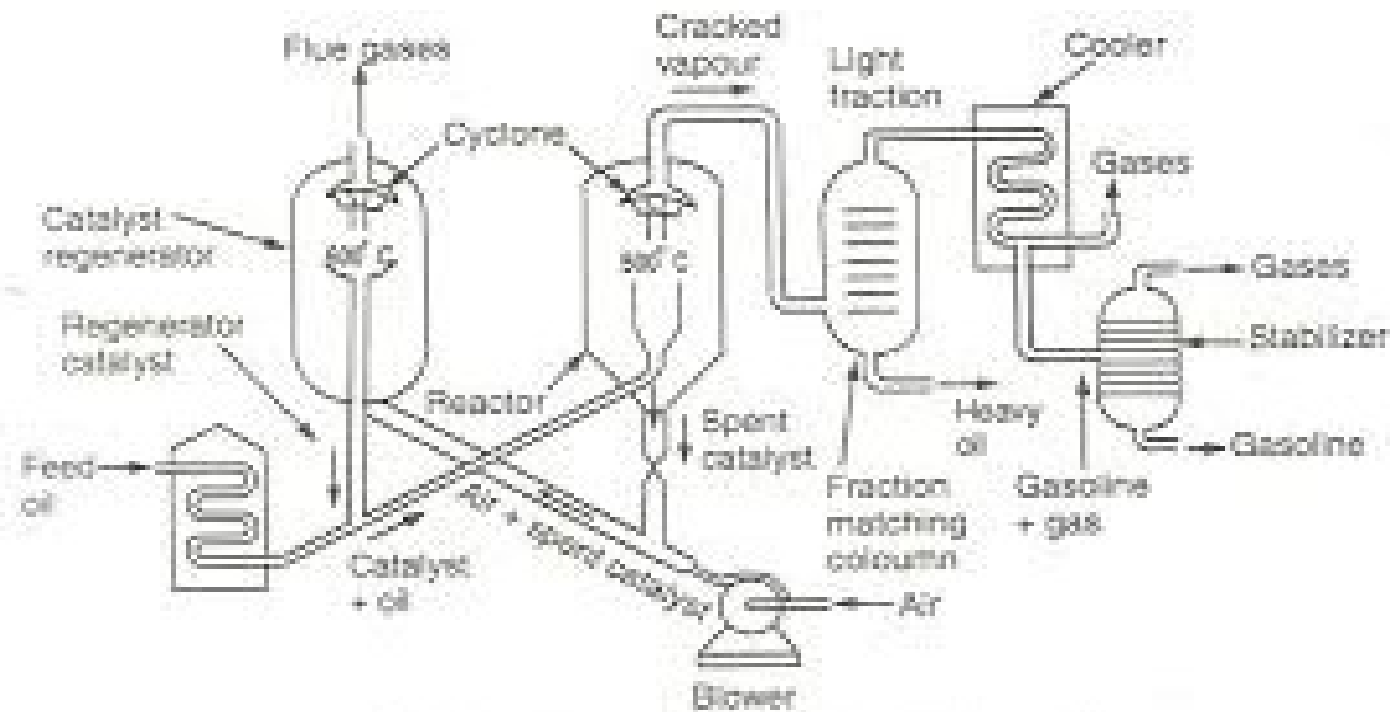


Fig. 9. Moving-bed type catalytic cracking.

MOVING-BED CRACKING

Advantages of Catalytic Cracking

1. The yield of petrol is higher.
2. The quality of petrol produced is better.
3. The cracking process can easily be controlled, so that desired products can be obtained.
4. A much lower pressure is needed in catalytic cracking.



Unit No: 1 Unit name:Applied Chemistry-1

Lecture No: 6

Knocking & Anti-knocking



Anti-knocking Agents

These are the substances added to petrol in order to prevent knocking in I.C. Engines.

Ex:

- i) TEL Tetra Ethyl lead.
- ii) TML Tetra Methyl lead.
- iii) MTBE Methyl Tertiary Butyl Ether.

Role of anti-knocking agents: They are added to the petrol to increase the octane number of the fuel which reduces the knocking. The process of adding anti-knocking agents to the fuel is called as doping.



Unleaded Petrol (ULP)

- Gasoline is normally mixed with 0.1% TEL to increase octane rating by 10-15 points. In order to overcome the harmful effects of TEL it is slowly phased out from gasoline. The petrol free from lead is called as unleaded petrol. It is now compulsory to use ULP for automobiles in several parts of our country due to the following reasons:
 1. Lead is toxic
 2. Lead reduces the efficiency of the catalyst used in converters.
 3. The petrol rating can be enhanced by the addition of certain compounds instead of TEL, Methyl-t- butyl ether (MTBE).



Unleaded Petrol (ULP)

Advantages of Unleaded Petrol

- It is fuel without Lead which is heavy metal & very harmful.
- The use of ULP helps in controlling knocking in engines as it supports the use of catalytic converters attached to exhaust in automobiles.
- ULP can also allows use of catalysts like Rhodium which converts exhaust gases like CO and NO to CO₂ & N₂. Rhodium also converts unburnt hydrocarbons to CO₂ & H₂O.



Classification of Crude Oil (Petroleum Oil)

The major classes of hydrocarbons in crude oil include the following:

1. Open chain alkanes
2. Cycloalkanes
3. Aromatics
4. Asphaltenes
5. Resins

1. Open chain alkanes- Both straight chain & branched chain alkanes are present in crude oil.



Classification of Crude Oil (Petroleum Oil)

2. Cycloalkanes – Crude oil contains cycloalkanes like cyclopentane & cyclohexane.

3. Aromatics - In all crude oils benzene & alkyl substituted benzenes are present.

4. Asphaltenes – The asphaltenes contain N, S, O etc.

5. Resins – These are the polymeric substances. They are gummy & are lower molecular weight polymers.



Unit No: 1 Unit name:Fuels

Lecture No: 7

Power Alcohol



Power alcohol

- When ethyl alcohol is used as fuel in internal combustion engine, it is called as "power alcohol". Generally ethyl alcohol is used as its 5-25% mixture with petrol.

Advantages of power alcohol

- Ethyl alcohol has good antiknocking property and its octane number is 90, while the octane number of petrol is about 65. Therefore, addition of ethyl alcohol increases the octane number of petrol.
- Alcohol has property of absorbing any traces of water if present in petrol.
- If specially designed engine with higher compression ratio is used, then disadvantage of lower calorific value of ethyl alcohol can be overcome.
- ethyl alcohol contains 'O' atoms, which helps for complete combustion of power alcohol and the polluting emissions of CO, hydrocarbon, particulates are reduced largely.
- Use of ethyl alcohol in petrol reduces our dependence on foreign countries for petrol and saves foreign considerably.
- Power alcohol is cheaper than petrol.

so why we are not using the ethyl alcohol because of following disadvantages.



-
- Disadvantages of power alcohol
 - Ethyl alcohol has calorific value 7000cal/gm much lower than calorific value of petrol 11500cal/gm. Use of power alcohol reduces power output upto 35%.
 - Ethyl alcohol has high surface tension and its atomisation, especially at lower temperature, is difficult causing starting trouble.
 - Ethyl alcohol may undergo oxidation reaction to form acetic acid, which corrodes engine parts.
 - As ethyl alcohol contains 'O' atoms, the amount of air require for complete combustion of power alochol is lesser and therefore carburettor and engine need to be modified, when only ethyl alcohol is used as fuel.

Use of ethyl alcohol in petrol reduces our dependence for petrol .



Unit No: 1 Unit name:Fuels

Lecture No: 8

Octane and Cetane number



Octane Number

- Measure of the ignition quality of gas (gasoline or petrol). Higher this number, the less susceptible is the gas to 'knocking' (explosion caused by its premature burning in the combustion chamber) when burnt in a standard (spark-ignition internal combustion) engine. Octane number denotes the percentage (by volume) of iso-octane (a type of octane) in a combustible mixture (containing iso-octane and normal-heptane) whose 'anti-knocking' [characteristics](#) match those of the gas being tested. In the older vehicles, high octane numbers were achieved by adding lead tetraethyl to the gas (the 'leaded gas'), a pollutant that contributes to lead poisoning (see Lead).
- In the newer vehicles, the same result is achieved by the engine design that increases turbulence in the combustion chamber, and/or by adding aromatic hydrocarbons (such as xylenes) and oxygenates (oxygen-containing compounds such as alcohols) to the gas (the 'unleaded gas'). Also called Octane rating. See also cetane number.



Cetane Number

- Measure of the ignition quality of diesel fuel; higher this number, the easier it is to start a standard (direct-injection) diesel engine. It denotes the percentage (by volume) of cetane (chemical name Hexadecane) in a combustible mixture (containing cetane and 1-methylnaphthalene) whose ignition characteristics match those of the diesel fuel being tested. See also Octane number.



Cetane Number



Unit No: 1 Unit name:Fuels

Lecture No: 9

Biodiesel



Biodiesel

- Chemically biodiesel is the methyl esters of long chain carboxylic acids. Biodiesel is obtained by transesterification of vegetable oil or animal fats with methyl alcohol using sodium metal or sodium methoxide as catalyst.

- **Transesterification**

It is the process of converting one ester into another ester. During the chemical conversion of vegetable oil to biodiesel we get water soluble glycerol & biodiesel.



Biodiesel

- The compounds present in biodiesel are like,
 1. Methyl palmitate : $\text{H}_3\text{C}-(\text{CH}_2)_{14}-\text{COOCH}_3$
 2. Methyl stearate : $\text{H}_3\text{C}-(\text{CH}_2)_{16}-\text{COOCH}_3$
 3. Methyl oleate : $\text{H}_3\text{C}-(\text{CH}_2)_7-\text{CH}=\text{CH}-(\text{CH}_2)_7-\text{COOCH}_3$
 4. Methyl linoleate : $\text{H}_3\text{C}(\text{CH}_2)_5(\text{CH}=\text{CH}_2)-(\text{CH}_2)_7-\text{COOCH}_3$



Biodiesel

Advantages of Biodiesel

1. It is cheaper.
2. It does not give out particulate & CO pollutants.
3. It is regenerative & environment friendly.
4. It is clean to use biodiesel in diesel engines.



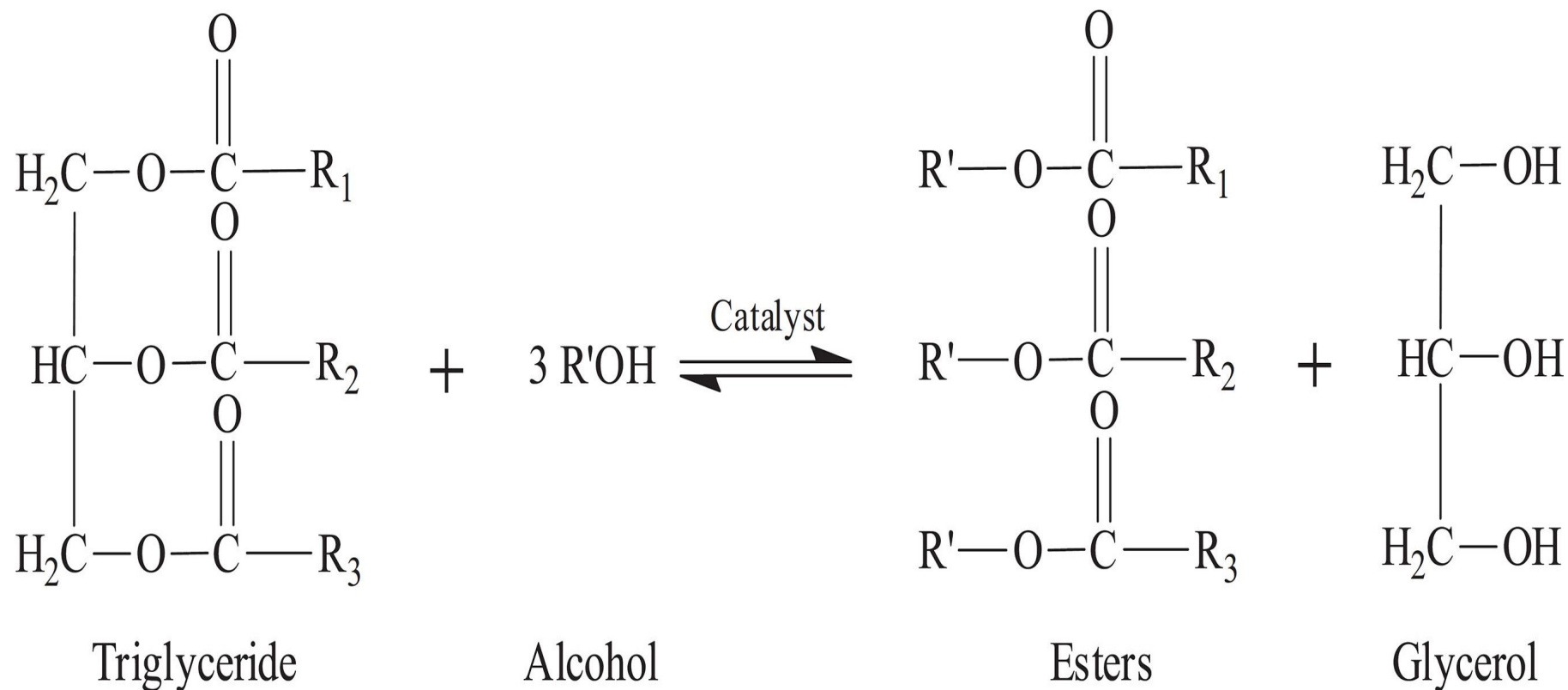
Biodiesel

Method / Steps to obtain Biodiesel- A Green Fuel

1. Filter the cheap or waste vegetable oil.
2. Heat it at 110c. With stirring to remove water from it.
3. Add the sodium methoxide about 2% by weight.
4. Then add methanol about 20%.
5. Heat the mixture with stirring for 30 minutes.
6. Cool & mix sufficient water, stir well. The glycerol & soap dissolved in water phase.
7. Separate the water insoluble phase (biodiesel) from water phase.



Biodiesel



Unit No: 1 Unit name:Fuels

Lecture No: 10

Propellant



Unleaded petrol

1. Unleaded petrol is one which does not contain any lead compound like ((TEL). To improve its octane no. concentration of high octane components like isooctane, isopentane , ethylbenzene , Isopropyl benzene etc. is increased by the process off reforming.
2. In addition to it, a compound like methyl tertiary butyl ether (MTBE) is added to improve the octane no. of unleaded petrol
3. This compound provides oxygen of ether group for combustion of petrol in I.C. engine, thereby reducing considerably the formation of peroxy compounds which causes knocking.



Advantages of Unleaded Petrol

Advantages of Unleaded Petrol

- Unleaded petrol permits the attachment of catalytic converters containing **rhodium catalyst** to the exhaust automobiles. The catalyst convert the toxic gases like **CO & NO** to non – toxic gases like **CO₂ & N₂** respectively. Consequently pollution level is reduced to great extent.
- The harmful effects of discharge of poisonous lead & its compound through exhaust of automobile is avoided.



Knocking

Knocking is defined as the sharp metallic sound produced in an internal combustion of engine result in loss of energy.

- It is related to the internal combustion (IC) of engine working on petrol. In IC engine, a mixture of gasoline vapours & air is used as a fuel.
- The combustion of the fuel is initiated by a spark in the cylinder. Due to combustion, gases are formed moving the piston down.
- The piston movement must be uniform without any vibration, but sometimes the combustion rate becomes so great that the fuel on ignition, suddenly increases the gas volume causing uneven movement of the piston with ratting noise in the engine which is a sharp metallic sound.



Knocking

Knocking causes :

- Loss of efficiency
- Damage to the piston & cylinder
- Loss of large amount of energy



Unit No: 1 Unit name:Fuels

Lecture No: 11

Unleaded Petrol



UNLEADED PETROL

Leaded Petrol: The petrol containing TEL or TML as anti knocking agents is called leaded petrol. TEL or TML are the very good anti knocking agents but has some disadvantages as follows.

- a) After combustion lead is deposited as lead oxide on piston and engine walls it leads to mechanical damage.
- b) Lead is a poisonous air pollutant..
- c) It spoils the catalyst used in catalytic converter.

Unleaded Petrol: The petrol, which contains anti-knocking agent other than lead, is known as unleaded petrol. Ex: **MTBE is used, as an antiknocking agent in place of TEL or TML and the petrol is known as unleaded petrol.**



Unit No: 1 Unit name:Fuels

Lecture No: 12

Combustion Reactions



Combustion

- Combustion is a process in which oxygen from the air reacts with the elements or compounds to give heat.
- As the elements or compounds combine in indefinite proportions with oxygen, we need to calculate what is minimum oxygen or air required for the complete combustion of compounds.



FUEL CELL

- A fuel cell is a device that converts the chemical energy from a fuel into electrical energy through a chemical reaction of positively charged hydrogen ions with oxygen or another oxidizing agents.
- A fuel cell is a special type of galvanic cell consisting of two electrodes & electrolytes that convert the chemical energy of a fuel (H_2 , Methanol etc.) & an oxidant (air & oxygen) into electricity.
- Fuel cell may be represented as:

Fuel / electrode / electrolyte / electrode / oxidant



TYPES OF FUEL CELL

A fuel cells can be divided into five major types based on the electrolyte used.

- Alkaline fuel cells (AFC)
- Phosphoric acid fuel cells (PAFC)
- Molten carbonate fuel cells (MCFC)
- Polymer electrolyte membrane fuel cells (PEMFC)
- Solid oxide fuel cells (SOFC)



FUEL CELL

Hydrogen – Oxygen Fuel Cell

- This is the simplest type of alkaline fuel cell. Here hydrogen is used as the fuel & oxygen gas is the oxidant.
- The cell consist of porous carbon electrode with catalyst Pt or Pd as anode.
- The cathode is also a porous carbon electrode with Pt or Ag as a catalyst.
- The electrolyte is an aqueous solution of KOH.



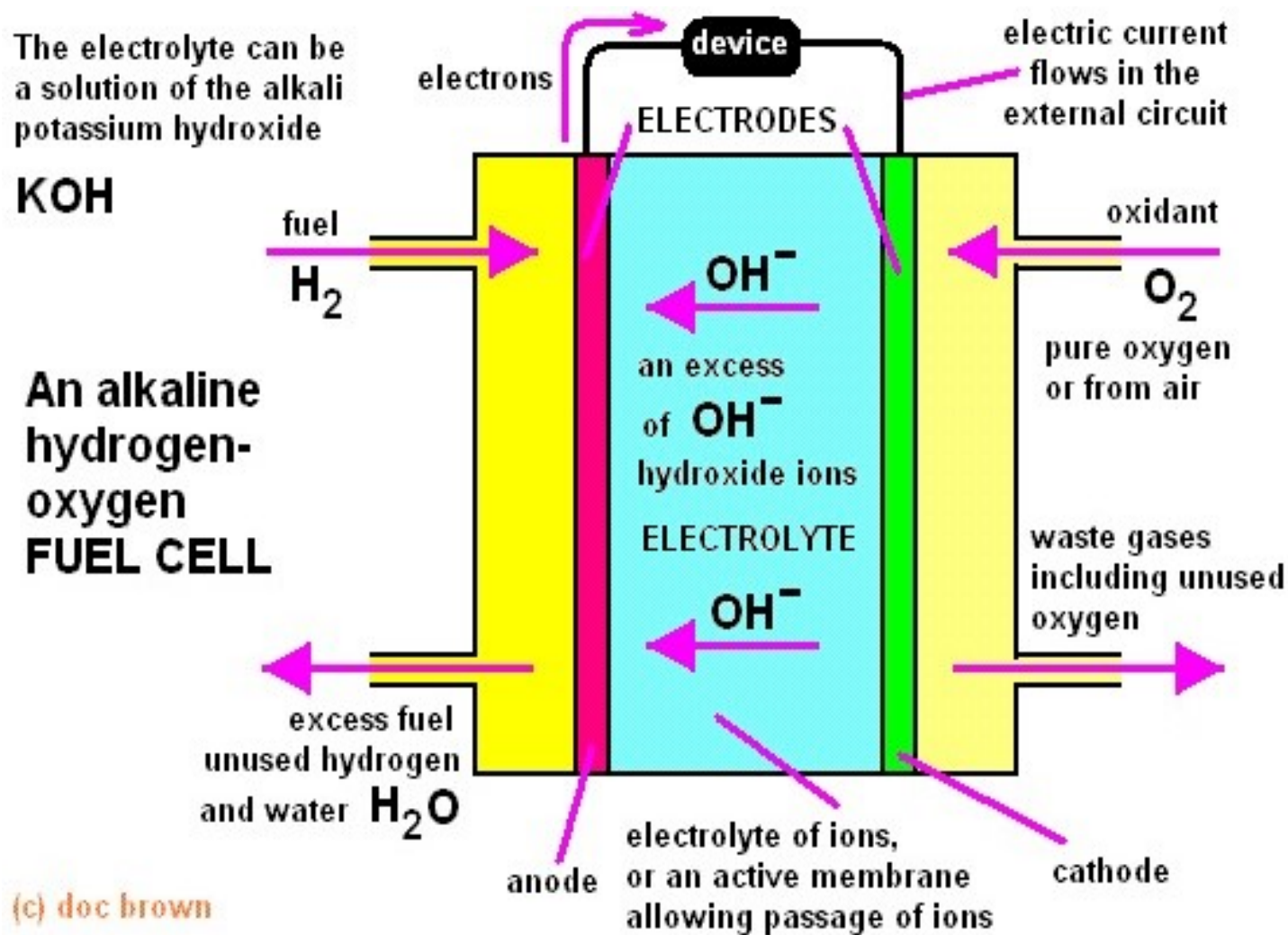
FUEL CELL

Hydrogen – Oxygen Fuel Cell

- The hydrogen gas fuel is continuously supplied at anode & oxygen gas at cathode.
- As the hydrogen gas diffuses through the anode, it is absorbed at the electrode surface & reacts with hydroxyl ions to form water
- At the cathode oxygen diffusing through the cathode is absorbed & reduced to hydroxyl ions



FUEL CELL



(c) doc brown



FUEL CELL

Hydrogen – Oxygen Fuel Cell

➤ The electrode reactions are as follows:



FUEL CELL

Advantages of Hydrogen – Oxygen Fuel Cells

- High efficiency of the conversion process
- Absence of harmful waste products.
- No moving parts & hence operates silently
- Requires less maintenance.



FUEL CELL

Disadvantages of Hydrogen – Oxygen Fuel Cells

- High initial cost.
- Power output is moderate.
- Fuels in the form of gases & O₂ need to be stored in tanks under pressure.



Thank You



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